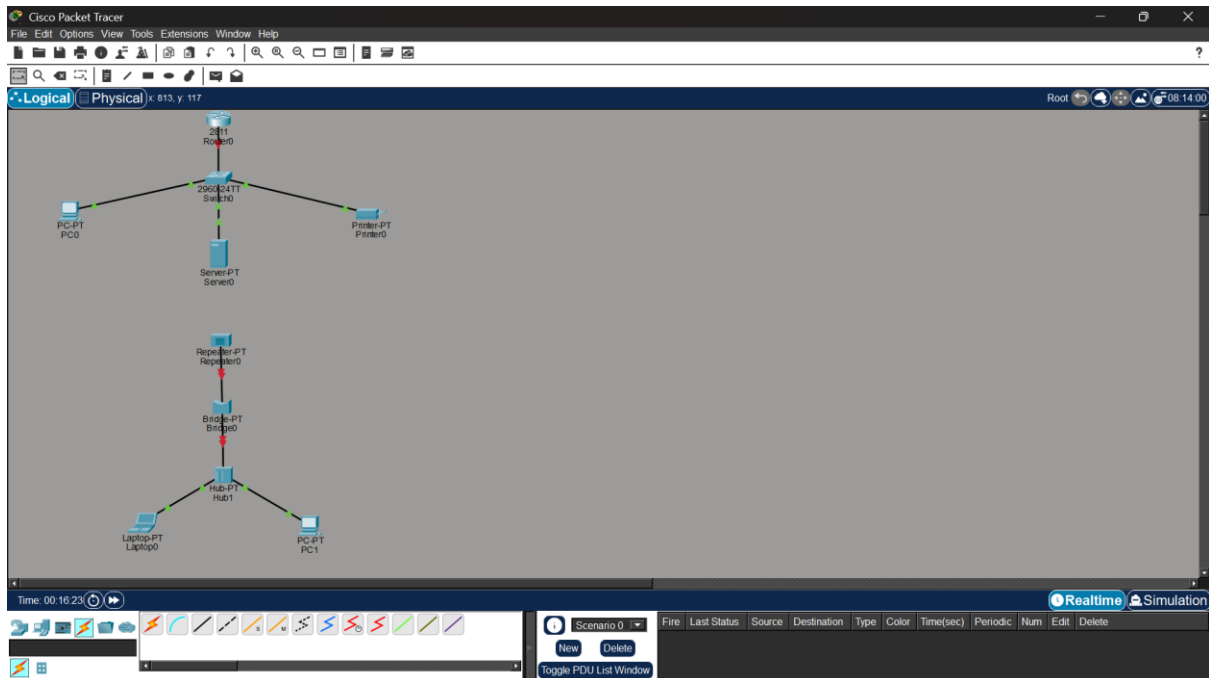
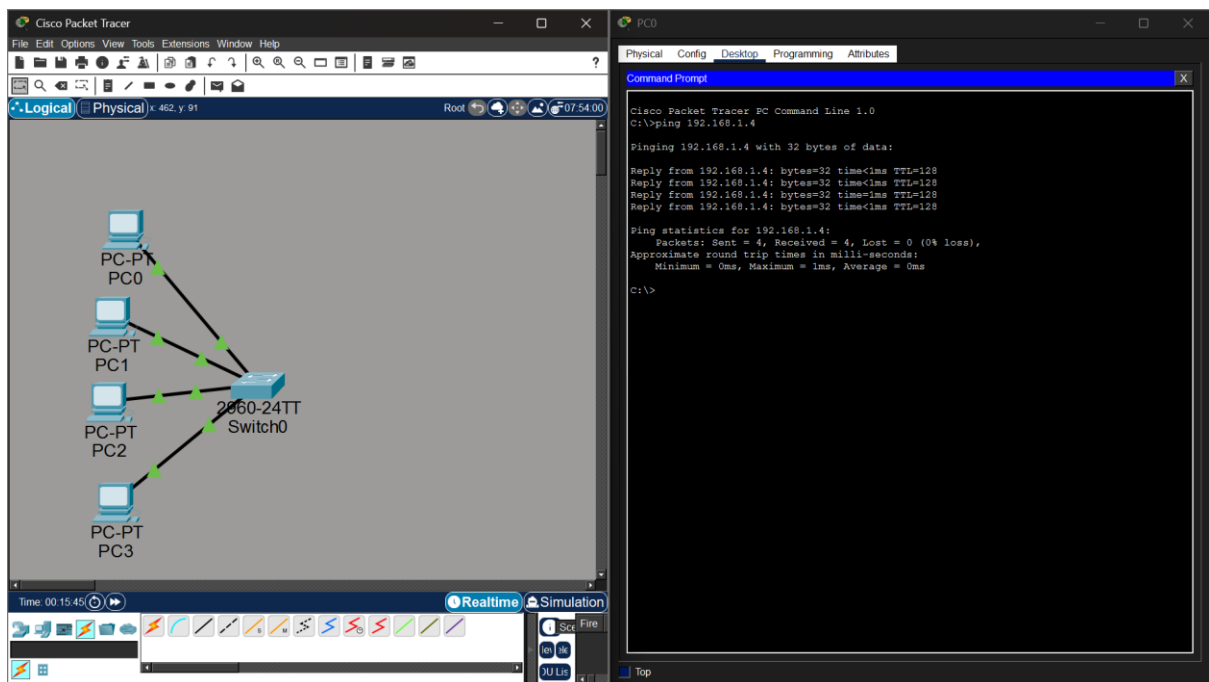


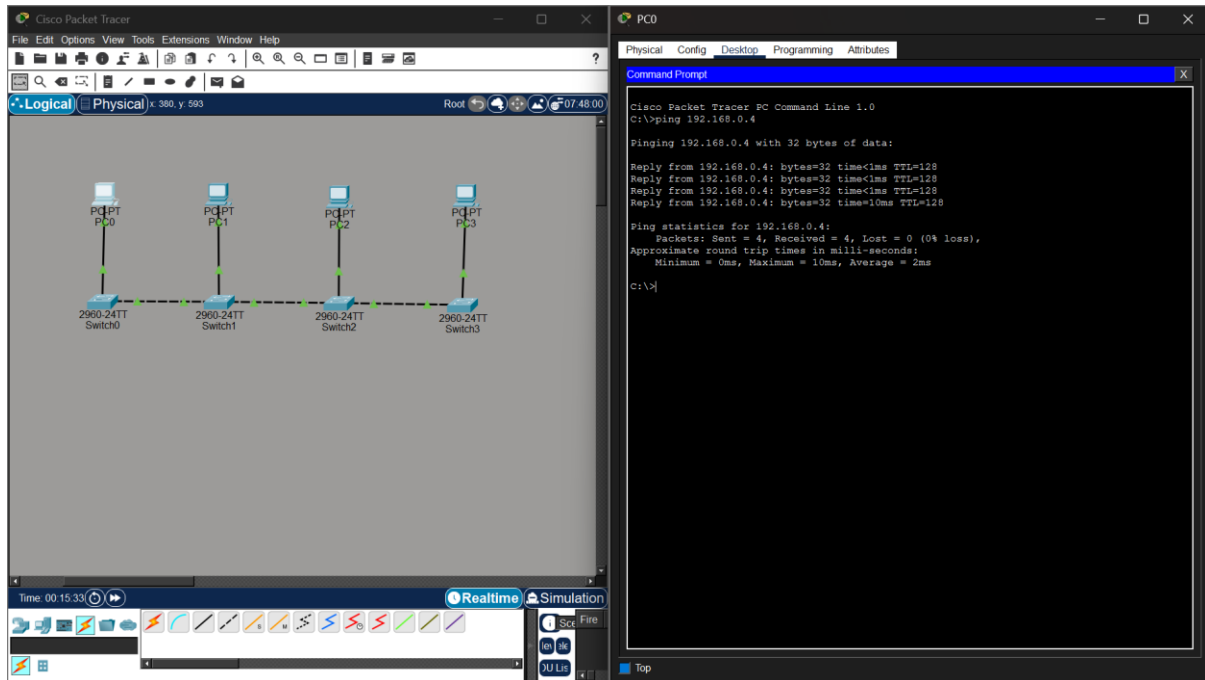
Experiment-01-Network-Components



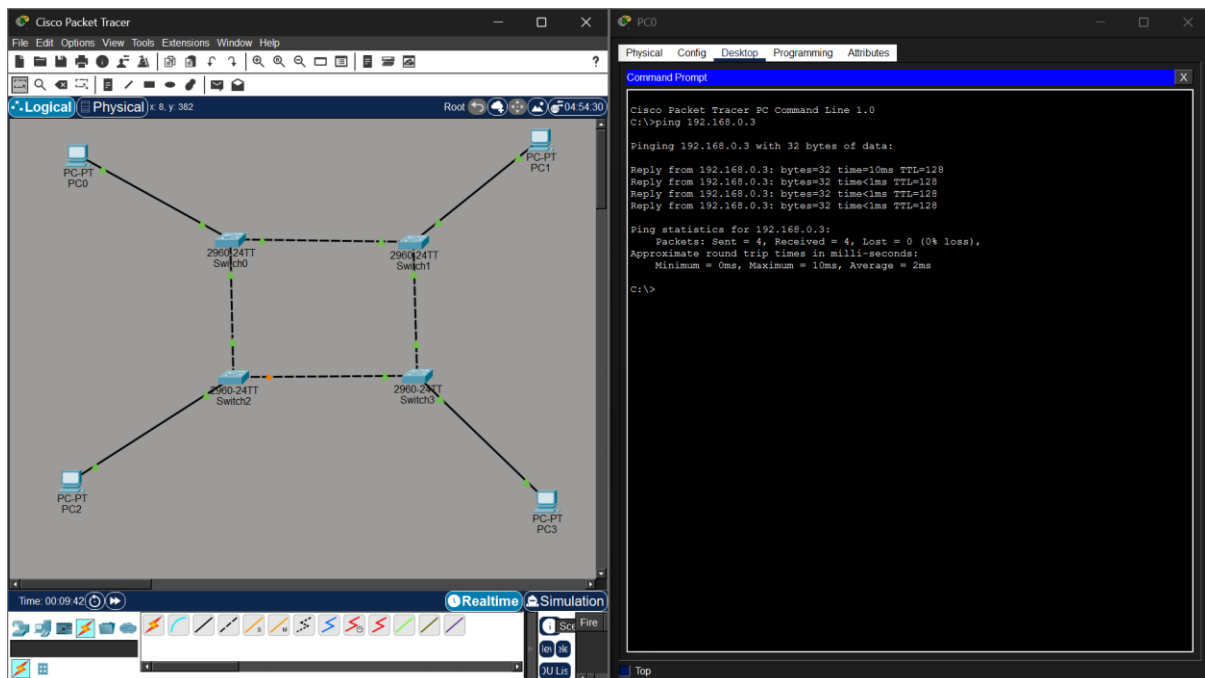
Experiment-02-Star-Topology



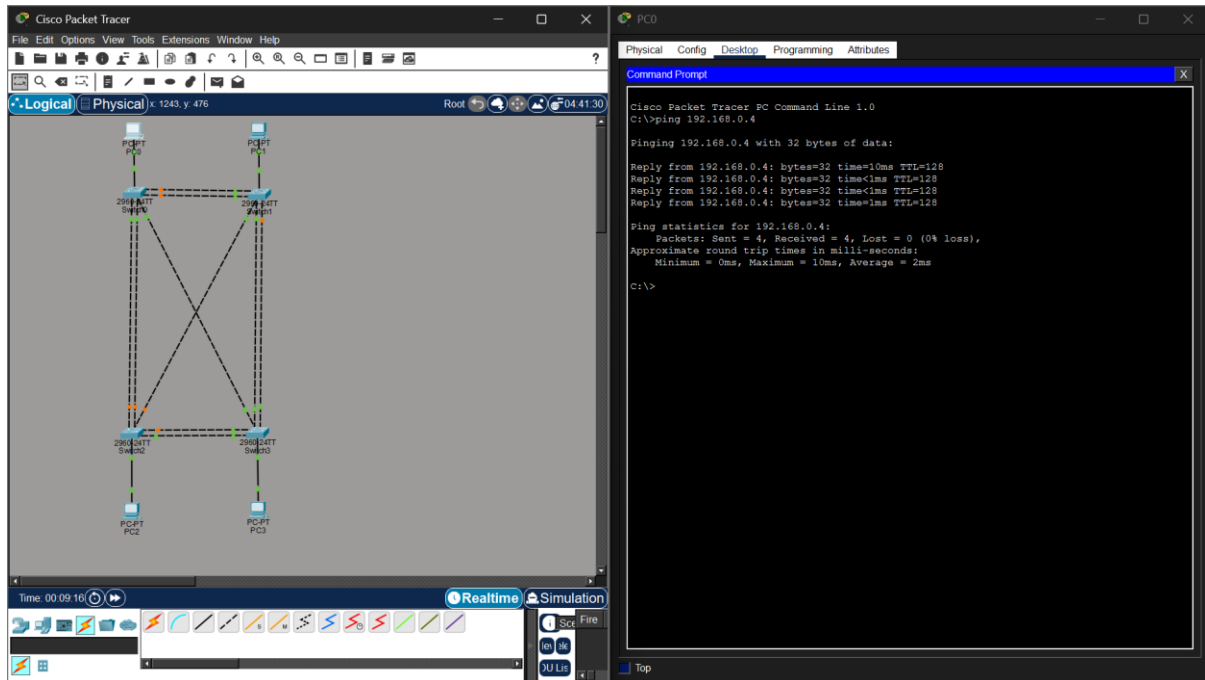
Experiment-03-Bus-Topology



Experiment-04-Ring-Topology



Experiment-05-Mesh-Topology



Experiment-06-Tree-Topology

Cisco Packet Tracer
File Edit Options View Tools Extensions Window Help

Logical Physical x: 1879, y: 983

Root 00:10:04:00

Simulation Panel

Event List

Vis.	Time(sec)	Last Device
0.005	Switch2	
0.005	PC3	
0.006	Switch2	
0.006	PC4	
0.006	Switch2	
0.007	PC5	
0.007	Switch2	
0.007	Switch2	
0.008	Switch2	
0.008	Switch0	
0.008	Switch1	
0.009	Switch0	
0.009	Switch1	
0.010	Switch1	
0.710	--	

Reset Simulation Constant Delay Capturing

Play Controls

Event List Filters - Visible Events

ACL Filter, Bluetooth, CAPWAP, CDP, DHCPv6, DTP, EAPOL, EIGRPv6, FTP, H.323, HSRPv6, HTTP, HTTPS, ICMP, ICMPv6, IEC, IPsec, ISAKMP, IoT, IoT, LACP, LLDP, MDIOBUS, Meraki, NDP, NETFLOW, NTP, OSPFv6, PAgP, POP3, PPP, PPPoE, PTP, Profinet, RADIUS, REP, RIPv2, RTP, SCP, SMTP, SNMP, SSH, SIP, SYSLOG, TACACS, TCP, TFTP, Telnet, UDP, USB, VTP

Edit Filters Show All/None

Event List Realtime Simulation

Time: 00:17:34.731 PLAY CONTROLS

Scenario 0

New Delete

Toggle PDU List Window

Cisco Packet Tracer
File Edit Options View Tools Extensions Window Help

Logical Physical x: 1539, y: 517

Root 00:07:36:00

Simulation Panel

Event List

Vis.	Time(sec)	Last Device
0.005	Switch2	
0.005	PC3	
0.006	Switch2	
0.006	PC4	
0.006	Switch2	
0.007	PC5	
0.007	Switch2	
0.007	Switch2	
0.008	Switch2	
0.008	Switch0	
0.008	Switch1	
0.009	Switch0	
0.009	Switch1	
0.010	Switch1	
0.710	--	

Reset Simulation Constant Delay Capturing

Play Controls

Event List Filters - Visible Events

ACL Filter, Bluetooth, CAPWAP, CDP, DHCPv6, DTP, EAPOL, EIGRPv6, FTP, H.323, HSRPv6, HTTP, HTTPS, ICMP, ICMPv6, IEC, IPsec, ISAKMP, IoT, IoT, LACP, LLDP, MDIOBUS, Meraki, NDP, NETFLOW, NTP, OSPFv6, PAgP, POP3, PPP, PPPoE, PTP, Profinet, RADIUS, REP, RIPv2, RTP, SCP, SMTP, SNMP, SSH, SIP, SYSLOG, TACACS, TCP, TFTP, Telnet, UDP, USB, VTP

Edit Filters Show All/None

Event List Realtime Simulation

Time: 00:14:57/ PLAY CONTROLS

Scenario 0

New Delete

Toggle PDU List Window

The top screenshot shows a Cisco Packet Tracer interface with a network topology. A central switch is connected to several PCs (PC0, PC1, PC2, PC3, PC4, PC5, PC6) and a server (PC-PT PC7). The interface includes a menu bar, a toolbar, and a simulation panel on the right. The simulation panel shows an event list with the following data:

Vis.	Time(sec)	Last Device
	0.000	--
	0.001	PC0
	0.002	Switch0
	0.003	Switch1
	0.004	PC3
	0.005	Switch1
	0.006	Switch0

The bottom screenshot shows the same network topology, but with the PDU List Window open. The PDU List Window displays a successful ICMP packet capture from PC0 to PC3. The interface also includes a menu bar, a toolbar, and a simulation panel on the right. The simulation panel shows an event list with the following data:

Vis.	Time(sec)	Last Device
	0.000	--
	0.001	PC0
	0.002	Switch0
	0.003	Switch1
	0.004	PC3
	0.005	Switch1
	0.006	Switch0

Experiment-08-ARP-Analysis

The image displays two screenshots of the Cisco Packet Tracer interface, illustrating a network setup and the process of analyzing ARP requests.

Top Screenshot: The network topology shows a central Hub0 connected to six PCs (PC-PT PC0 through PC-PT PC5). The PDU List Window is open, showing a successful ICMP packet from PC0 to PC3. The simulation is running in Realtime mode.

Bottom Screenshot: The network topology is the same, but the PDU List Window shows a successful ICMP packet from PC0 to PC3. The Event List Panel is open, showing a list of events. The simulation is running in Realtime mode.

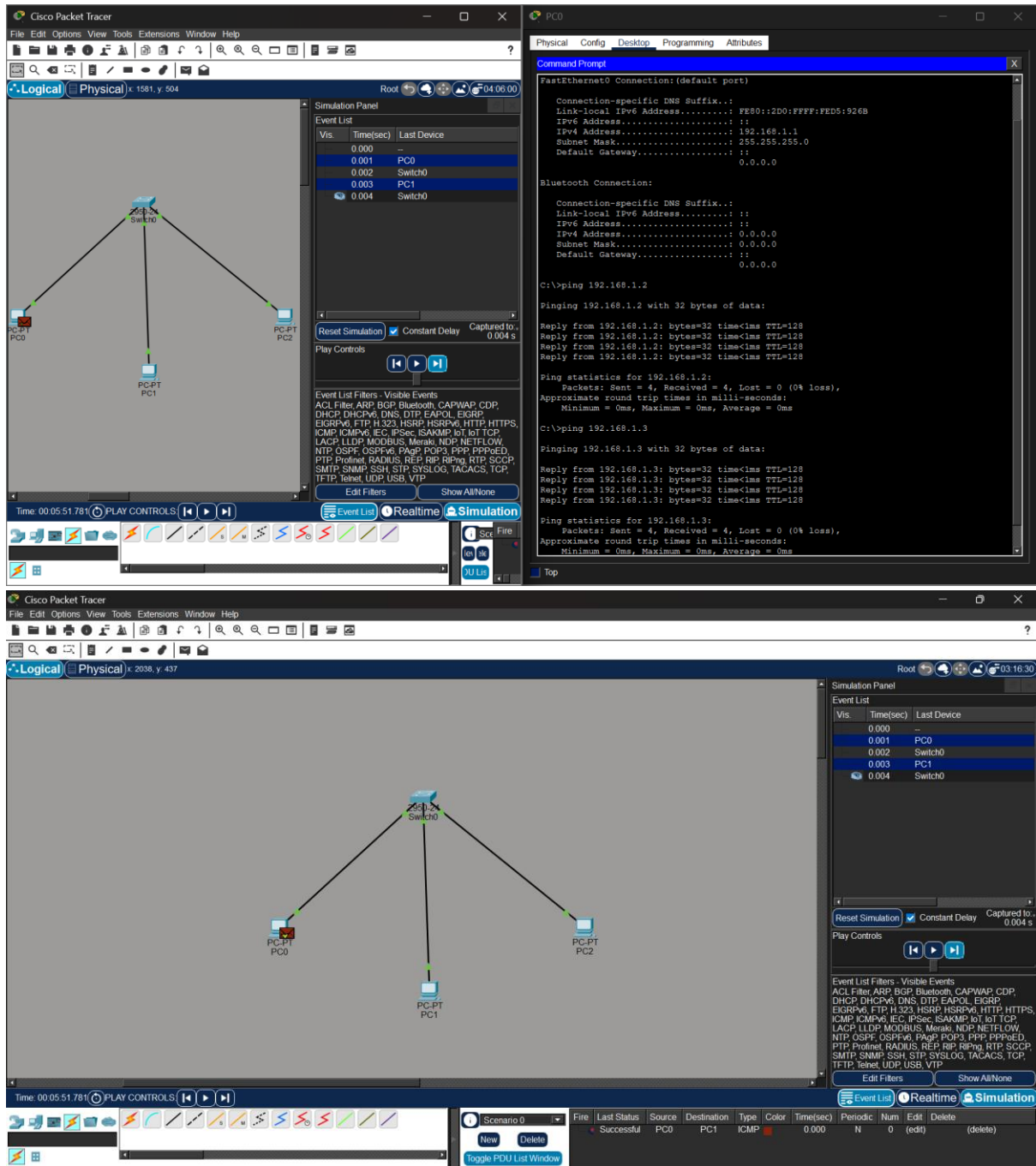
Event List Panel:

Vis.	Time(sec)	Last Device
	0.000	
	0.001	PC0
	0.002	Hub0
	0.002	Hub0
	0.002	Hub0
	0.002	Hub0
	0.003	PC3
	0.004	Hub0
	0.004	Hub0
	0.004	Hub0
	0.004	Hub0

PDU List Window:

File	Last Status	Source	Destination	Type	Color	Time(sec)	Periodic	Num	Edit	Delete
	Successful	PC0	PC3	ICMP		0.000	N	0	(edit)	(delete)

Experiment-09-CSMA-CD-CSMA-CA



Experiment-10-Computer-Lab

The image displays two screenshots of the Cisco Packet Tracer software interface. The top screenshot shows a network topology with a central switch (2960-24 Switch0) connected to four PCs (PC-PT PC0, PC-PT PC1, PC-PT PC2, PC-PT PC3) and a router (1841 Router0). The bottom screenshot shows the same network topology, but with a Command Prompt window open on PC0, displaying the results of a ping command to 192.168.1.254.

Top Screenshot: Network Topology

- Router: 1841 Router0
- Switch: 2960-24 Switch0
- PCs: PC-PT PC0, PC-PT PC1, PC-PT PC2, PC-PT PC3

Bottom Screenshot: Command Prompt Output

```
C:\>ipconfig

FastEthernet0 Connection (default port)

    Connection-specific DNS Suffix...:
    Link-local IPv6 Address . . . . .: FE80::230:F2FF:FETA:2C39
    IPv6 Address . . . . .:
    IPv4 Address . . . . .: 192.168.1.1
    Subnet Mask . . . . .: 255.255.255.0
    Default Gateway . . . . .:

Bluetooth Connection:

    Connection-specific DNS Suffix...:
    Link-local IPv6 Address . . . . .:
    IPv6 Address . . . . .:
    IPv4 Address . . . . .: 0.0.0.0
    Subnet Mask . . . . .: 0.0.0.0
    Default Gateway . . . . .:

C:\>ping 192.168.1.254

Pinging 192.168.1.254 with 32 bytes of data:

Reply from 192.168.1.254: bytes=32 time=12ms TTL=255
Reply from 192.168.1.254: bytes=32 time<1ms TTL=255
Reply from 192.168.1.254: bytes=32 time<1ms TTL=255
Reply from 192.168.1.254: bytes=32 time<1ms TTL=255

Ping statistics for 192.168.1.254:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 12ms, Average = 3ms

C:\>ping 192.168.1.4

Pinging 192.168.1.4 with 32 bytes of data:

Reply from 192.168.1.4: bytes=32 time=1ms TTL=128
Reply from 192.168.1.4: bytes=32 time<1ms TTL=128
Reply from 192.168.1.4: bytes=32 time<1ms TTL=128
Reply from 192.168.1.4: bytes=32 time<1ms TTL=128
```